A Deep Learning Approach to Fabric Defect Detection: CNN and ResNet for Efficient Quality Assessment

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***Abstract*—In the textile industries, the detection of defective fabrics is one of the most important tasks associated with quality control. Traditional methods used for defect detection are time-consuming and include human errors. This paper, therefore, presents an automatic fabric defect detection model based on Dee techniques using CNN and ResNet for higher accuracy in detecting defects in big-sized textile samples. It includes image preprocessing, feature extraction, and defect classification. To improve model generalization, 48,000 labeled images in six defect classes are augmented with rotations. Performance of these architectures is analyzed, and advantages of ResNet deeper architecture with transfer learning capability are discussed. The proposed model automates the inspection process, with a significantly quicker process that decreases costs when benchmarked against the traditional method. In this way, textile processing quality control will be smarter and more effective.**

Keywords— fabric defect detection, quality control, CNN, ResNet, deep learning, image preprocessing, defect classification, transfer learning

I. Introduction

The textile industry is a crucial industry that provides high-quality fabrics to meet very divergent consumer demands. It remains one of the oldest and most vital sectors that has traditionally depended on manual techniques in ensuring the quality of fabrics. However, as the demands by consumers increase with increasing scales of production, the inadequacies in manual inspections become more conspicuous. This process, besides being naturally very well open to human error and being duly manual, becomes less effective when large lots of fabrics are to be processed. Quality assessment of fabric is extremely important since defects such as holes, stains, and other abnormalities may make a big difference in the finished product. Conventionally, defect detection in textiles has been performed manually, which is a very time-consuming job and prone to human mistakes. This tends not only to be costly but also an inefficient way, particularly when volumes of fabric are large [1].

The above-mentioned problems can be treated with the help of a model that is capable of automatically detecting defects in fabrics using more advanced deep learning techniques. More specifically, our approach applies CNNs and ResNet to detect flaws within large textile samples with high accuracy. [2].

In this paper, we will introduce a deep learning-based model incorporating CNNs and ResNet for automatic fabric defect detection. It intends to replace traditional manual inspections with a much automated, scalable solution capable of processing huge volumes of fabrics faster and with little human intervention. Advanced techniques, such as image preprocessing, feature extraction, and defect classification, are integrated into our model to provide the best framework for fabric quality control [3].

II. Literature Review

One of the most challenging tasks in the textile industry regarding quality control concerns the detection of fabric defects [4]. The following will chronologically review some of the key studies in fabric defect detection in order to show how work in this area has evolved.

2024: YOLOv8 and MobileNetSSD FPN Lite on Real-World Manufacturing Data: In 2024, a study focused on real-world scenarios, applying models such as YOLOv8 and MobileNetSSD FPN Lite is designed to work in manufacturing data for defect detection of plain fabrics [5]. YOLOv8 topped with a high score of 84.8%, while difficult to handle noisy samples and fabrics, which is patterned; this means the need for more robust models that can best fit multiple types of fabrics and industry environments.

2023: Cascade R-CNN, SAC, Custom Dataset: The approach taken in the year 2023 was more advanced by using Cascade R-CNN, which incorporated SAC on a custom dataset of 1,604 images spanning nine defect categories. The mean average precision achieved was 75.33% [6]. However, significant limitations were with regard to small dataset size and a high computation burden. Another recent study, from 2023, focused on fabric defect detection and compared ResNet with VGG-16 models, where the best performances given by VGG-16 were 73.91% accuracy versus ResNet, which reached an accuracy of 67.59% [7]. This work mentioned that the model performed weakly when it needed to detect several defects in one image.

2021: CNNs, SVMs, and Eigenfilters on TILDA and MVTec AD Datasets: A 2021 study explored the performance of deep learning models, including CNNs, SVMs, and eigen filters, on the TILDA and MVTec Anomaly Detection datasets [8]. While the automatic modes of these models achieved 75%, operator-assisted modes allowed performance increases up to 95%. That was a study which pioneered deeper model exploration for higher autonomy and accuracy.

2019: Pretrained CNNs and Defect Segmentation: In the year 2019, some pre-trained CNNs were applied, which gave an accuracy of 97.31%. This study focused on underlying defect segmentations of prime importance. This paper indeed gave an established result increase as opposed to the previous works; hence, it underlined the efficiency of transfer learning in troubleshooting tasks of fabric defects [9].

Overall, the domain of fabric defect detection goes incrementally from typical image processing techniques to deep learning models. However, challenges include real-time performance, dataset limitations, handling of multiple defects, and robustness against a noisy environment.

III. Methodologies

CNN and ResNet have been used in this study to enhance the accuracy of fabric defect detection. Preprocessing steps include resizing, normalizing, and patch-based segmentation with a focus on defect localization. Data is divided into training and testing sets, with batch processing implemented for efficient memory use. Features are extracted via CNN/ResNet, then classified using a fully connected layer. Model training is guided by cross-entropy loss and optimized with Adam.

a. Dataset

Our dataset comprises 48,000 labeled images representing various fabric segments, with image dimensions of either 32x32 or 64x64 pixels to capture a range of detail and complexity in defect identification[11]. The images are categorized into six distinct defect classes, as outlined below:

Table 1: Distinct Defect Classes

Defect Class	Description
Good	Defect-free fabric, serving as a baseline for comparison.
Color	Issues in fabric coloration due to dye inconsistencies or fading.
Cut	Cuts or tears in the fabric affecting its integrity.
Hole	Holes that significantly compromise fabric usability.
Thread	Loose or improperly woven threads indicating a flaw in the manufacturing process.
Metal Contamination	Presence of metal particles, which can impact both fabric quality and safety.

b. Preprocessing

Powerful preprocessing in turn affirms the reliability and strength of our fabric defect detection model. Our preprocessing pipeline consisted of three main stages-image processing, text processing for defect labels, and feature integration, as illustrated in Figures 1, 2, and 3 respectively. They calibrate and enhance the dataset, hence establishing solid grounds for efficient model training and results you can impudently stand up for.

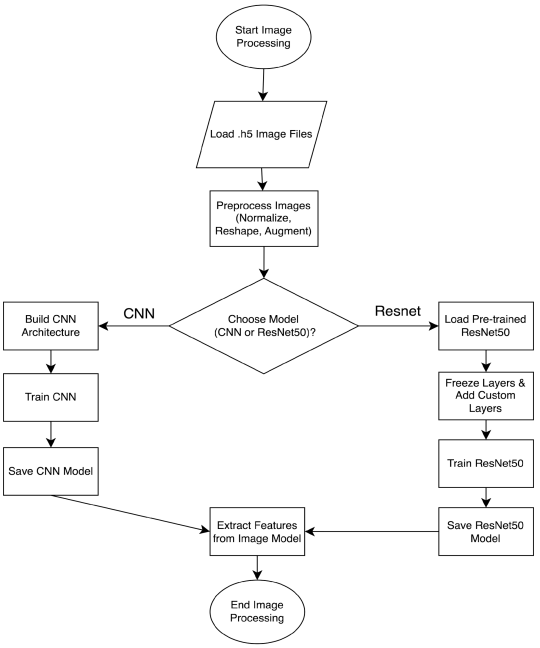


Figure 1: Image Preprocessing

The image processing workflow (Figure 1) begins by loading .h5 image files, each representing a segment of fabric with a label indicating its defect type. Images thus preprocessed to a dimension of 32x32 would suit the specifications for a model. The color values are subjected to normalization between 0 and 1, which stabilizes the process and leads to faster convergence on the training set [8]. Also introduced are data augmentations: transformations like rotation (0°, 20°, and 40°), flip, and random cropping to improve either generalization by the model or to reduce overfitting [12]. Each image must be interpreted and indexed into sub-blocks, allowing for the model to only focus on small portions, consequently facilitating detection of minute defects or isolated ones without taxing deployment on hardware resources.

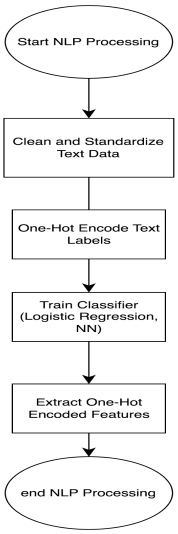


Figure 2: NLP Processing

The image processing functions parallel with textual labels, which are a separate preprocessing workflow representing defect classes (Figure 2). Label normalization starts with standardizing all the labels first, and is then subjected to an encoding that has one-hot categories for translating categorical text labels into numerical vectors [13]. In addition, auxiliary classifiers, such as logistic regression, should also be trained on these exhausted labels to validate the process of label encoding and for it to be consistent with the downstream model requirements [14].

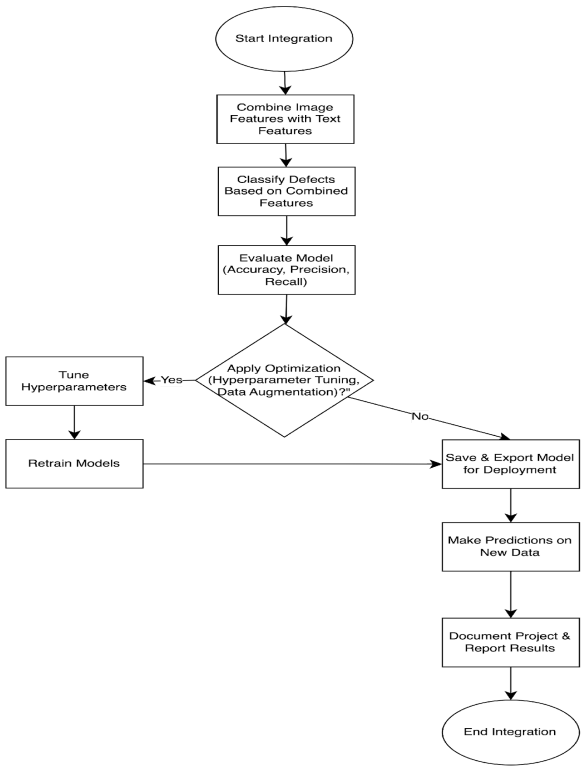


Figure 3: *Integration*

In the integration stage (Figure 3), image features extracted from CNN/ResNet models are combined with encoded text features to create a comprehensive dataset for defect classification. This combined feature set leverages both visual and categorical information, improving the model's ability to classify defects accurately [15] . Metrics for evaluation include accuracy, precision, and recall, after which the model underwent iterative optimization through hyperparameter tuning and augmentation if deemed necessary. The model then achieved optimal deployment performance, after which it is saved and exported for mass use on new data and predictions. A thorough documentation of the preprocessing steps, model performance, and final configuration is the basis of reproducibility and its continual improvement.

The integrated structure of the preprocessing pipeline is to ensure that the model will accommodate quite a few diverse types of defects in real-world textile samples, thereby augmenting both the accuracy and scalability of automated systems in fabric quality control.

c. Model Selection

Our model selection strategy incorporates Convolutional Neural Networks (CNN) for grayscale images and Residual Networks (ResNet) for RGB images, optimizing defect detection across varied fabric types and defect categories. MaxPool-2D layers allow for the down-sampling of feature maps to get spatial information with low computation. They are followed by the Flatten layer, which converts these 2D matrices to one-dimensional vectors that are passed through Dense layers and softmax layers until this last stage retroactively categorizes the input in one of the earlier specified defect classes [16].

To improve generalizability and possibly reduce overfitting, a Dropout layer is introduced in the fully connected layers that randomly deactivated neurons during training [17]. The CNN model was compiled with the Adam optimizer, which uses an adaptive learning rate, and trained using a sparse categorical cross-entropy loss function, which is suitable for multi-class classification tasks. Training is performed for a total of 200 epochs with a batch size of 64, thus balancing computational efficiency with memory occupancy. A validation split of 20% enables real-time monitoring of the model's performance on unseen data, facilitating early detection of overfitting. Both accuracy and loss metrics are tracked for training and validation, providing insights into the model's learning progression and classification capability.

For RGB fabric images, a ResNet which is fine-tuned by transfer learning employing pretrained weights from large-scale datasets. That helps convergence quicker while enhancing performance, this happens thanks to features learned before generalization [18]. A GlobalAveragePooling2D layer has been added to the architecture to resize the ResNet feature maps into a compact representation of input features. The characteristics of this approach are the preservation of information pertinent to the image while reducing the chances for overfitting. It employs a hybrid architecture of the unifying CNN for the grayscale image and ResNet for color images. It enables the identification of a wide range of defects which enhances robustness and versatility. The adaptable multi-model framework is applicable for a variety of fabric types and defect scenarios.

IV. Implementation

In this study, we implemented two deep learning models- Convolutional Neural Network and Residual Network to classify images in the dataset. The choice of these two models was based on their respective strengths.

A. CNN Model

The CNN model has an input shape of (32, 32, 1) obtained by reshaping the 32x32 grayscale images in order to form a single channel layer. The first convolutional layer training involved 64 filters, all of which were (3x3) and swept across the total image to produce a variety of feature maps. The ReLU activation function is also employed in this convolution. After convolution, the max pooling layer with the size of (2 x 2) is used to make a sample of the previous layer. The second convolutional layer escalates the amount of filters to 128 allowing the network to extract more elaborate features. Another max pool layer is used to further down sample the spatial dimensions. The approach enables the model to construct a hierarchy of features going from simple to more complex. The above convolutional layers the output signal is flattened into a 1D vector. This vector is further taken through fully connected dense layers, where the last layers of the network's decisions are made. To reduce the problem of overfitting a dropout layer is added before the last layer of the network.

Training and Testing: The CNN model was trained using the Adam optimizer to achieve an optimal convergence rate [22]. Training of the model was performed up to 200 epochs with a batch size of 128 and the training data were divided in 80% for training dataset and 20% for validation. While training, the accuracy and loss of the proposed model architecture were measured with respect to the training data as well as the validation data. Both early stopping techniques, where training would be ceased if for some epochs the validation loss does not decrease, were used to avoid the model from overfitting. In the classification level, the CNN model improved the accuracy significantly during the training phase and also during the validation phase. In the final testing, the accuracy of the CNN model was retrieved at 72%-76% indicating its capacity to address the complexity of the given data set despite the need to fine tune it more.

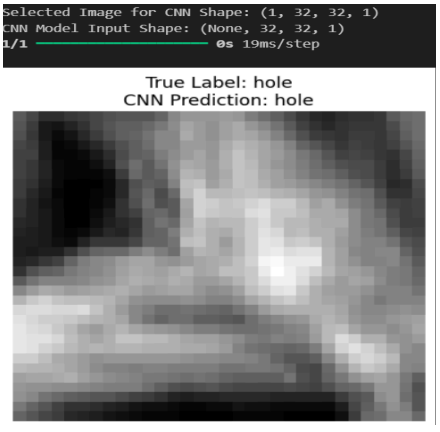


Figure 4: Prediction of True Label

B. Resnet Model

The ResNet was specifically constructed for addressing the vanishing gradient problem which is more frequent in the deep networks and normally symptomatic of gradients that are too small to pass through the network during backpropagation [17] [21]. The model was pre-trained on the ImageNet dataset, and transfer learning was applied to adapt this powerful model to our specific classification task.

The ResNet model employed is truncated from the pre-trained convolutional layers of ResNet [17]. The pre-trained layers act as a feature extractor, trying to find important traits within the image based on the vast information trained into the ImageNet database. Since it requires RGB images as input, the grayscale images in the dataset were converted to RGB. This resulted in images of shape (32, 32, 3), which are suitable for the ResNet architecture. The convolutional layers in ResNet progressively extract features from the images, moving from lower-level features like edges and textures to higher-level semantic features such as shapes and objects.

Training and Testing: For the CNN model of ResNet, the Adam optimizer and sparse categorical cross entropy loss were used in the course of model training.This was done by freezing the pre-trained layers. Resnet training does not allow the training algorithm to change them, but the new layers added on top were trained to prevent them from being modified, while the custom layers added on top were fine-tuned. It included the training for 150-200 epochs and the size of the batches was equal to 128. This allowed the model to retain the powerful feature extraction capabilities while adapting to the specific task of classifying images in the current dataset [22]. Training was conducted for 150-200 epochs with a batch size of 128. CNN model used a validation split of 20% to evaluate the model's performance during training. A confusion matrix was generated to visualize the classification accuracy across different classes.

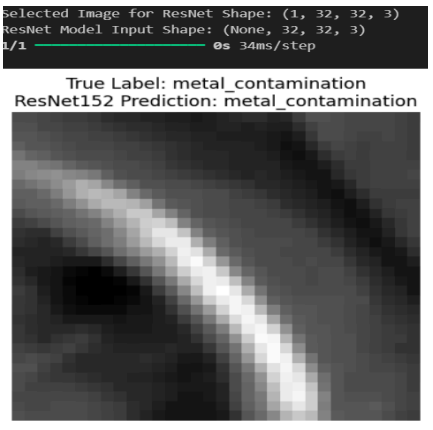


Figure 5: Prediction of True Label

V. Comparison Of Models

Both the CNN and ResNet models were trained and evaluated on the same dataset, allowing for a direct comparison of their performance. The CNN model demonstrated an accuracy of 76%. This result was expected given the relatively shallow architecture of the CNN and its focus on spatial hierarchies. However, the model required more training epochs to converge and

exhibited some limitations in distinguishing between visually similar classes.

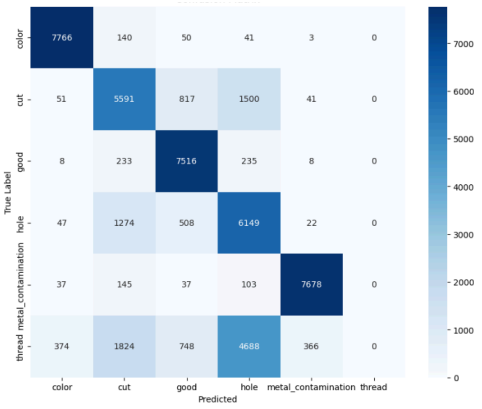


Figure 5: CNN Confusion Matrix

The ResNet models were expected to offer improved accuracy of 72%. The depth of the ResNet architecture allowed it to capture more complex features, particularly those that were difficult for CNN to identify. However, since only the top layers were fine-tuned, its success depended on how well these layers could adapt the pre-trained features to the specific task at hand.

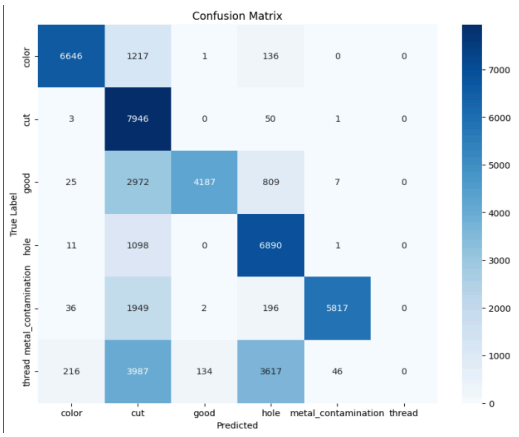


Figure 5: Resnet 152 Confusion Matrix

In conclusion, while both models performed adequately, the deeper ResNet architecture with transfer learning potentially provided more robust feature extraction, though this is subject to further evaluation of its final accuracy. The comparison between CNN and ResNet in this study underscores the importance of selecting the appropriate architecture based on the complexity of the dataset.

VI. Results

The CNN model achieved a final testing accuracy of 76%, demonstrating its ability to handle the complexity of the dataset, while the ResNet model achieved 72%. This performance gap highlights the benefits of ResNet's deeper architecture and transfer learning capabilities. The performance of the two models, CNN and ResNet, was compared based on their final testing accuracy. The confusion matrix for each model provides a detailed analysis of classification accuracy across different defect classes, identifying areas where the models excel and where they encounter challenges.

VII. Future Scope

- Real-time Performance: Further research is needed to optimize the models for real-time defect detection, crucial for high-throughput industrial applications.
- Handling Multiple Defects: The models should be further developed to handle multiple defects within a single image, a common scenario in textile production.
- Robustness to Noise: The models' robustness to noise and variations in fabric texture needs to be improved for reliable performance in real-world industrial settings.
- Integration with Production Lines: Integration of the model with existing production lines and automation systems is crucial for seamless implementation and optimization of the defect detection process.

VIII. Contribution

In this project we used CNNs for grayscale images and ResNet for RGB images, which improved the identification of different kinds of defects. The most significant part of our intervention, therefore, was identifying the general and specific performance of the model through the process of comparing the predicted labels with the actual labels. It helped us compare the results and define the accuracy of the classification of defects and what changes can be made.

IX. Conclusion

This research demonstrates the feasibility and effectiveness of deep learning models for automated fabric defect detection. The proposed model, utilizing CNN and ResNet architectures, significantly enhances the accuracy and efficiency of defect classification compared to traditional manual inspection methods. It emphasizes the benefits of ResNet's deeper architecture and transfer learning capabilities, along with the importance of dataset augmentation for model robustness. This automated approach holds the potential to revolutionize fabric inspection in the textile industry, paving the way for smarter and more efficient quality control systems that can meet the demands of modern manufacturing.

X. References

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